

Aaryan Shakya

AI Engineer · Machine Learning & Deep Learning

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PROFESSIONAL SUMMARY

AI Engineering student with hands-on experience building and training ML/DL models and neural networks, including cleaning and transforming raw data into model-ready datasets. Experienced in building retrieval-augmented generation (RAG) systems and processing large-scale datasets, with a strong interest in applying these skills in an AI/ML workflow.

CORE TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, React, HTML/CSS, C, C++

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, NumPy, Pandas, Polars, Dask, GeoPandas

Methodologies: Graph Neural Networks (GNNs), LLMs & Prompt Engineering, Retrieval-Augmented Generation (RAG), Computer Vision (CV), ETL/ELT Pipelines

Tools & Dev Workflow: Git, Docker, n8n (workflow automation), AI coding assistants & agentic IDEs

KEY TECHNICAL PROJECTS

Spatial Commercial Suitability Modeling — Kathmandu Valley *Geospatial ML*

Built a multi-label classifier predicting 6 commercial categories across Kathmandu using Overture Maps data on Uber's H3 spatial grid; engineered multi-scale proximity/POI features via R-Tree joins and trained MLP & Balanced Random Forest models with per-class threshold tuning.

LEGIT — Legal Expertise through Generative & Extractive Pre-trained Transformers *NLP / LLM*

Built a 10M-parameter language model from scratch for a low-resource Nepali legal corpus (~4M tokens) with a custom Devanagari OCR-cleanup pipeline; trained a 6-layer transformer decoder achieving 0.6432 validation loss, 79.4% next-token accuracy, and 1.9 perplexity.

EmerG-Net — Discovering Growing Communities in Graphs *GNN / Research Infra*

Designed a framework flagging early-stage research clusters before citation saturation by fusing 768-d SciBERT embeddings with node(paper) embeddings; built the FastAPI backend for incremental graph updates.

Audio Emotion Recognition via Adaptive Graph Neural Networks *GNN*

Reframed speech emotion recognition as graph learning, modeling audio channels as nodes with adaptive relational structure (TodyNet / dyGIN2d) to classify speech into 7 emotional states.

Enterprise Workflow Automation & Agentic RAG Systems *LangChain / n8n*

Built event-driven automation workflows connecting multiple APIs, including an LLM-backed query transformation pipeline using function calling and structured output parsing.

Multimodal Nepali Cultural Dance Classification Engine *Multimodal ML*

Built a multimodal neural network classifying 20 traditional Nepalese dance styles, aligning features across multiple sensory modes.

RESEARCH & PUBLICATIONS

Deep Reinforcement Learning for Asymmetric Strategy in Baghchal (Co-author)

Benchmarked MuZero, PPO, DQN & REINFORCE on the asymmetric board game Baghchal in a custom PyTorch environment; MuZero's Tiger agent reached an 86% win rate, PPO was most resource-efficient.

Nepalese Currency Counterfeit Detection System (Co-author)

Co-built a CV system verifying NPR 500/1000 banknotes across 7 security features; published an open-source 300+ image dataset and combined ORB region extraction with SSIM scoring.

EDUCATION

B.Tech. in Artificial Intelligence

Kathmandu University, Dhulikhel, Nepal

Expected Graduation: 2027

LEADERSHIP & MENTORSHIP

Organizing Committee & Mentor | AI Convergence 2026 (KUAIC): Mentored university students through an intensive 3-day data camp covering automated data wrangling with Pandas, statistical visualization with Matplotlib, and introductory ML pipelines.

Hackathon Competitor | KU x Chunjae Education Bootcamp: Completed a rigorous 10-day applied AI sprint and 3-day hackathon, engineering functional machine learning proofs-of-concept under strict time constraints.

LANGUAGES

English: Fluent (Professional Working Proficiency)

Nepali: Native/Bilingual Proficiency